Jacobs Building Redesign

MAE Enterprise

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Head of Finance Malek Gallichio
Head of Design Elizabeth Vazquez

EG 1003 Section A2 | Milestone 3 Presentation | April 8, 2021

Agenda

- Project Objective
- Background Information
- Technical Design Description
- Cost Estimate
- Project Schedule
- Teamwork Agreement
- Summary

Project Objective

Mission Statement

At MAE Enterprise, not only do we construct buildings with top-performing security, accessibility, and necessities, we make a second home for students and faculty.

Figure 1: MAE Enterprise logo

MAE Enterprise has been requested to redesign the old Jacobs building that:

- Has new classrooms
  - Ten 40-student classrooms
  - Eight 20-student classrooms
  - Two 100-student classrooms
- One recreational space
- Dorm space for 100 students
- One facility of choice
  - Theater

Project Objective

- Design with Autodesk Revit 2019, a computer-aided design (CAD) tool.
- Organize floors logically
  - Lower floors: welcoming spaces, exhibitions, administrative offices
  - Middle floors: classrooms
  - Upper floors: dorms and facilities to accommodate students
- Take advantage of the old Jacobs building location
  - Location and Transportation: parking lot
  - Sustainable Sites: park space

Background Information

MAE Enterprise has been commissioned to redesign the old Jacobs Administrative Building by New York University as a college campus extension to the Tandon School of Engineering. The lot is 200’ x 200’.

- Goals
  - Satisfy client
  - Spark intrigue to the Brooklyn campus
  - Increase number of incoming students
  - Get recommended to other clients

Technical Design Description

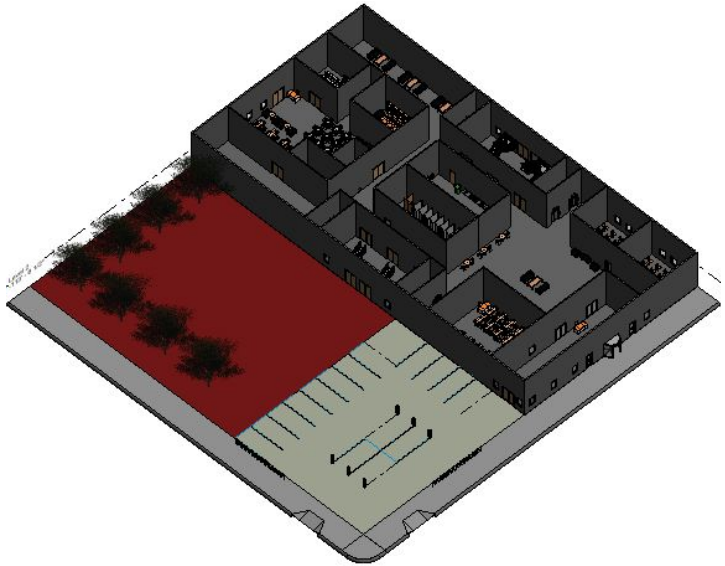


Figure 2: Previous 3D View

Since Milestone 2, we have:

- Added components outlined in the project objective
- Completed necessary electrical and plumbing plans

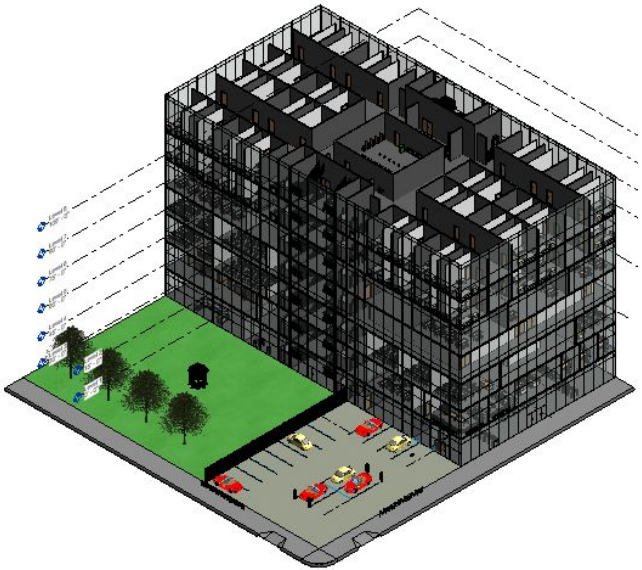


Figure 3: Current 3D View

Technical Design Description

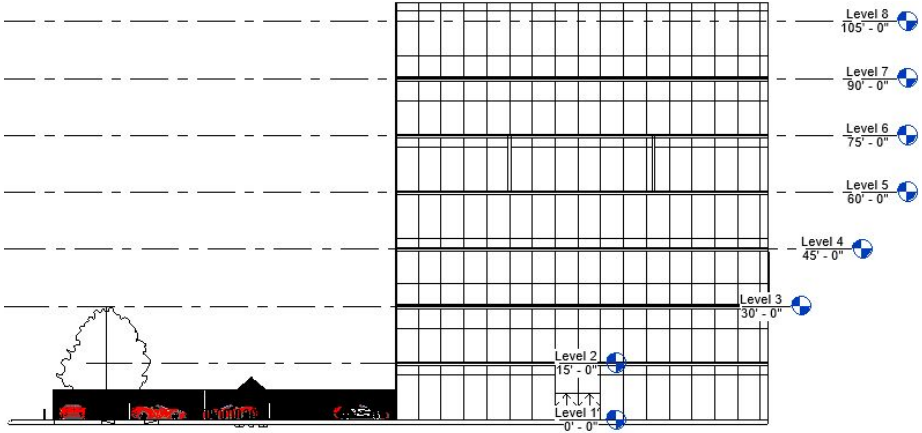


Figure 4: Front elevation

Figure 5: Most detailed side elevation

Technical Design Description

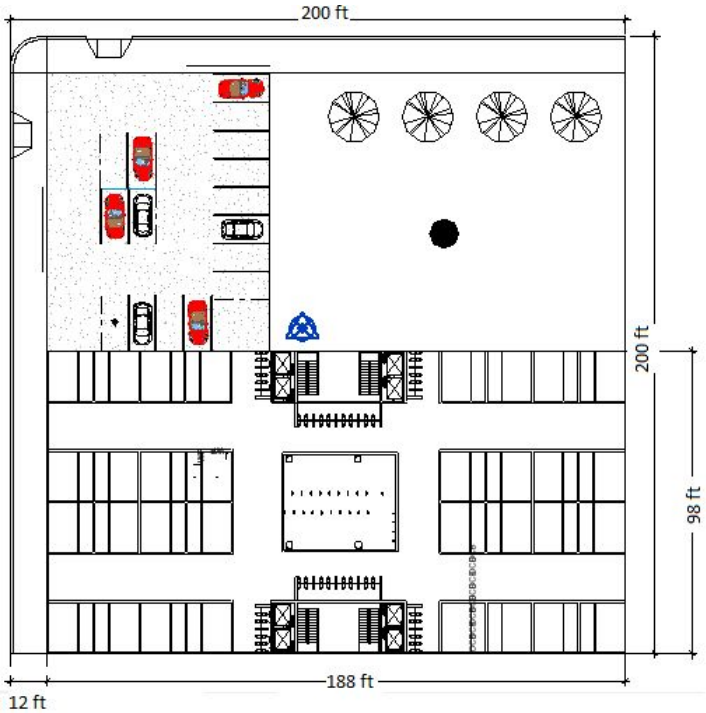


Figure 6: Site plan

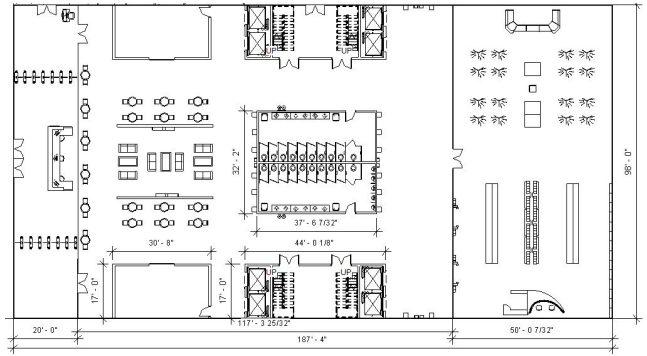


Figure 7: First Floor Plan

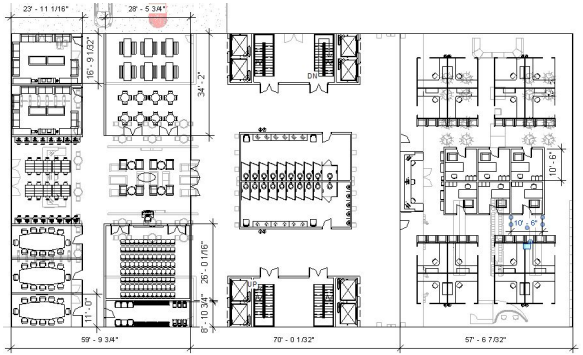


Figure 8: Second Floor Plan

Technical Design Description

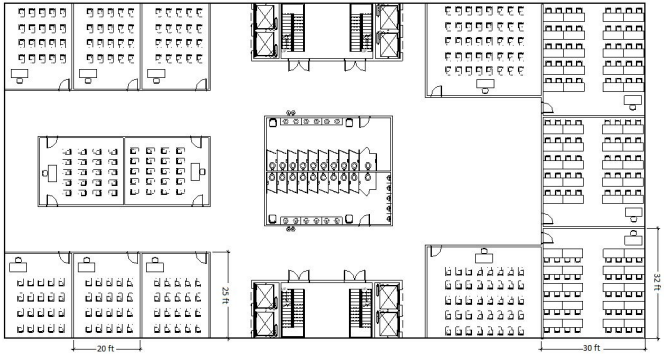


Figure 9: Third Floor Plan

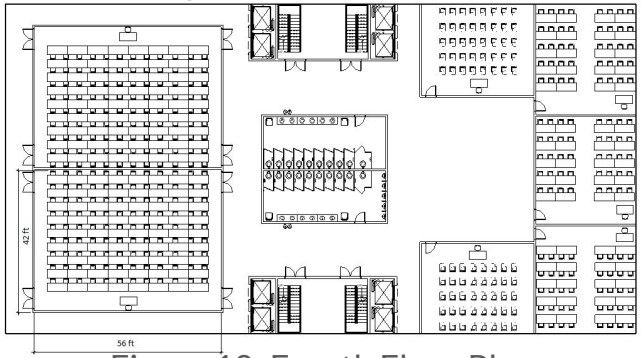


Figure 10: Fourth Floor Plan

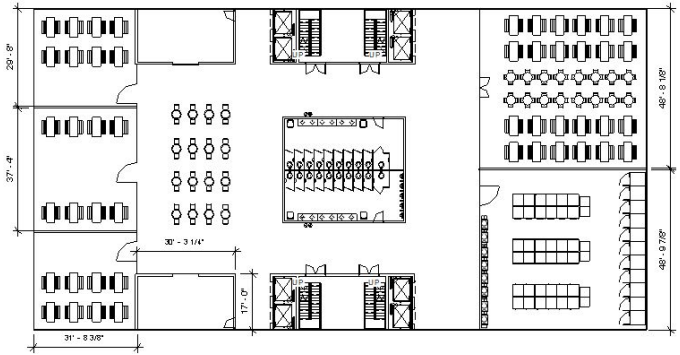


Figure 11: Fifth Floor Plan

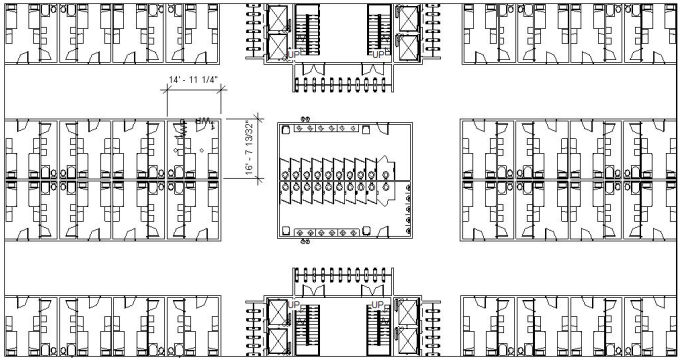


Figure 12: Sixth and Seventh Floor Plan

Technical Design Description

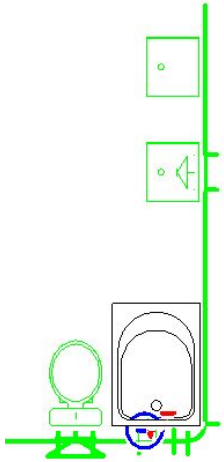


Figure 13:
Plumbing plan
for dorm
bathroom

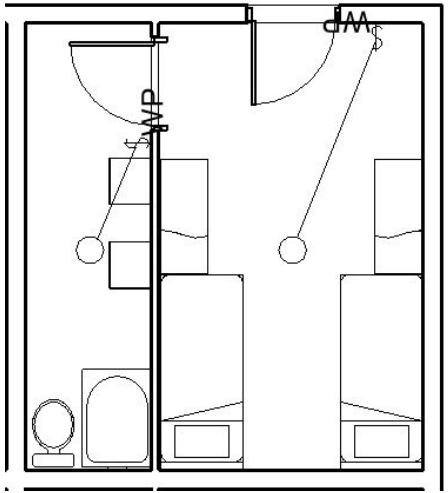


Figure 14:
Electric plan for
dorm &
bathroom

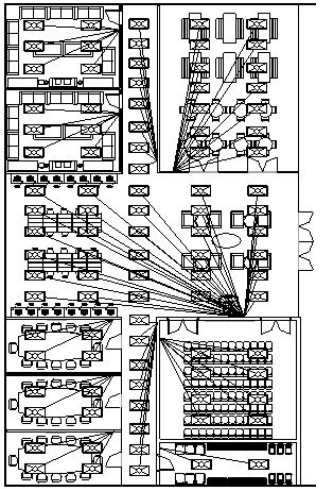


Figure 15:
Electric plan for
recreational
center

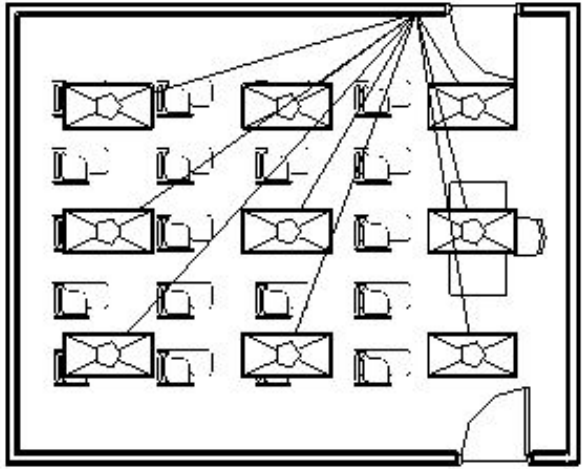


Figure 16:
Electric Plan for
one classroom

Cost Estimate

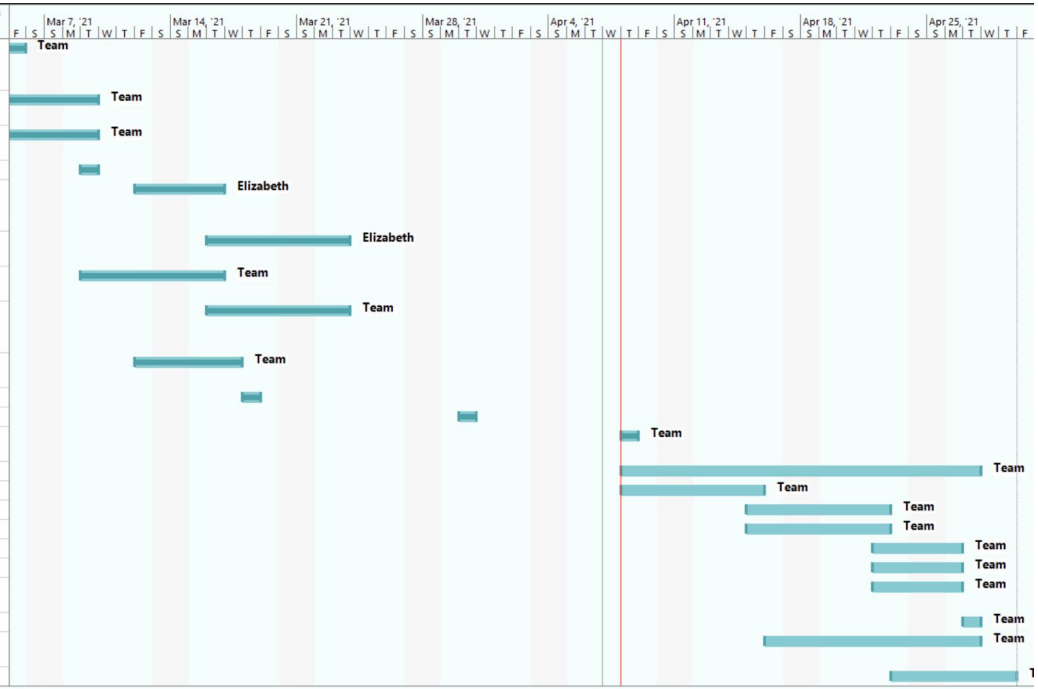
Table 1: Cost Estimate

Category	Material	Unit	Units Needed	Cost/Unit	Total Cost
Foundation	Steel	sq/ft	240,000	$9.75	$2,340,000
	Concrete	sq/ft	240,000	$5	$1,200,000
	Bricks	unit	2700000	$0.55	$1,485,000
Electrical	Electrical System	sq/ft	240000	$8	$1,920,000
	Security System	n/a	1	$20,000	$20,000
	Elevators	unit	8	$35,000	$280,000
Finishing	Thermal Protection	sq/ft	40800	$1.19	$48,552
	Windows	units	190	$150	$28,500
	Paint	gallons	156	$30	$4,680
	Furniture	n/a	1	$100,000	$100,000
	Educational Supplies	n/a	1	$500,000	$500,000
Miscellaneous	Miscellaneous	n/a	1	$1,500,000	$1,500,000
Labor	Manual Labor (50)	hours	200000	$50	$10,000,000
	Head of Project Management	hours	2000	$85	$170,000
	Head of Finance	hours	1500	$80	$120,000
	Head of Design	hours	2500	$60	$150,000
	Engineer (3)	hours	9000	$65	$585,000
				Total Cost	$20,451,732
				Total Cost + %20	$24,542,078

Project Schedule

Table 2: Project Schedule

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	✓		Discuss LEED Categories and plan Design	1 day	Fri 3/5/21	Fri 3/5/21		Team
2	✓		Design Student Classrooms	3 days	Fri 3/5/21	Tue 3/9/21		Team
3	✓		Design and 3D print logo	3 days	Fri 3/5/21	Tue 3/9/21		Team
4	✓		Benchmark A	1 day	Tue 3/9/21	Tue 3/9/21		
5	✓		All Student Classrooms Revit Design	3 days	Fri 3/12/21	Tue 3/16/21		Elizabeth
6	✓		One Recreational Space Revit Deisgn	6 days	Tue 3/16/21	Tue 3/23/21		Elizabeth
7	✓		Facility of choice revit design	6 days	Tue 3/9/21	Tue 3/16/21		Team
8	✓		Electrical and Plumbing for Dorm Room	6 days	Tue 3/16/21	Tue 3/23/21		Team
9	✓		Milestone 2 Presentation	4 days	Fri 3/12/21	Wed 3/17/21		Team
10	✓		Milestone 2	1 day	Thu 3/18/21	Thu 3/18/21		
11	✓		Benchmark B	1 day	Tue 3/30/21	Tue 3/30/21		
12	✓		Milestone 3 Presentation	1 day	Thu 4/8/21	Thu 4/8/21		Team
13			Revit Drawings	14 days	Thu 4/8/21	Tue 4/27/21		Team
14			All Floor Plans	6 days	Thu 4/8/21	Thu 4/15/21		Team
15			Plumbing System	6 days	Thu 4/15/21	Thu 4/22/21		Team
16			Electrical System	6 days	Thu 4/15/21	Thu 4/22/21		Team
17			Front Elevation	3 days	Thu 4/22/21	Mon 4/26/2		Team
18			Most Detailed Side	3 days	Thu 4/22/21	Mon 4/26/2		Team
19			Model of Building (3D Revit)	3 days	Thu 4/22/21	Mon 4/26/21		Team
20			Comissioning	1 day	Tue 4/27/21	Tue 4/27/21		Team
21			LEED Accredition Report	8 days	Fri 4/16/21	Tue 4/27/21		Team
22			Final Presentation	5 days	Fri 4/23/21	Thu 4/29/21		Team



Summary

- On schedule
  - Finished Revit designs and drawings
- On budget
  - Used the materials needed
- Move forward with commissioning and LEED accreditation report
  - Location and Transportation
    - Parking lot
  - Sustainable Sites
    - Park space
  - Water Efficiency
    - Plumbing fixtures & indoor plumbing
  - Indoor Environmental Quality
    - Vents

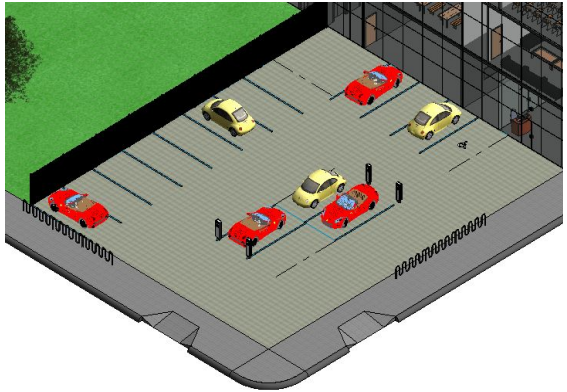


Figure 17: Parking Lot

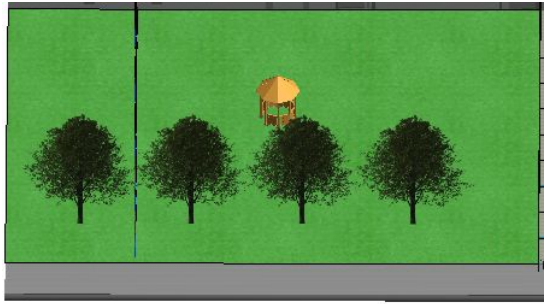


Figure 18: Park Space

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Head of Design Elizabeth Vazquez

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